

Diploma in Data
Analysis

Introduction to Data Analysis

Lesson 1: Summary Notes



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Lesson Objectives

- **Objective 1:** Introduction to data analysis
- **Objective 2:** Introduction to data
- **Objective 3:** Importing and cleaning data

Lesson Introduction

The future is rapidly evolving into one that is incredibly data driven, where decisions are made based upon data analysis. It is said that the amount of data in the world doubles every 2 years as more information becomes available. One of the major challenges we face considering all this data, is how to extract useful insights from it and how we can use this data to make better, more informed, and accurate decisions. One of the ways that we can better utilise this data is through data analysis. Data analysis helps make sense of this mass amount of data by extracting useful insights and interpreting them in a meaningful way. At the end of the day, everyone can benefit from learning more about analysing data!



Introduction to Data Analysis

Data analysis is the process of investigating, cleaning, transforming, and modeling the colossal volume of data that is available today. Why do you want to do this? To extract useful information and make more informed decisions!

Statistician John Tukey defined data analysis in 1961 as: “Procedures for analysing data, techniques for interpreting the results of such procedures, ways of planning the gathering of data to make its analysis easier, more precise or more accurate, and all the machinery and results of (mathematical) statistics which apply to analysing data” (Tukey and Cleveland, 1986).

The data analysis journey

Data analysis can be looked at as a journey, where with each step you get to know the data a little bit better.

Introduction to the problem

Here, we might be asked by a business to help with an issue they face, or we might have a question about a problem, and we want to use data to solve that problem.

- We want to understand what the stakeholder needs are in this step of the journey.
- What needs to happen before you can dive into the data?
- We need to understand what stakeholders want from data.
- We need to understand the question that the stakeholders are posing to us. Ways of doing this would be to ask specific questions, for example: Which channel should we focus on more to raise revenue?

Remember: It is your responsibility to mitigate what you are capable of and what is possible with the data available and how long it will take. It should not take months to analyse, but important to give yourself enough time to avoid mistakes. Set realistic expectations with the data available and how much time it will take to analyse the data.

Prioritise

Now that we are aware of the problem, we need to decide which of these questions need answering are the most important.

- Identify what the questions are that you need to answer, what are your goals? I always like to create a list of my tasks and then assess the importance and urgency of each.
- How would the stakeholder measure the value of each question they need to answer?
- Are there any tasks that are quick wins? (i.e. something that is easily ‘ticked off the list’ that will need minimal effort.

Open communication about the process with the stakeholder is key, so be open to any changes they might want.

Which tool should you use?

Once you have decided upon which question to tackle first, you need to decide on which tools will help you achieve your goals.

There are many different tools to utilise for data analysis out there today. It is always better to master one before moving onto another, but by understanding the complexity of data analysis you want to undertake, you can choose the tool best suited for your needs.

During this course, you will learn more about data analysis through tools **Excel, R and Tableau**.

Excel

Excel traditional tool for analysing data and where we will start our journey.

In this course, I assume that you have some prior experience with Excel, but if you have never been exposed to this tool, rather head to Shaw Academy's Excel course first to get better acquainted with this tool.

Excel is a great tool for analysing data, especially if you want employees of various technical skills to analyse data, but it is just one of the many tools that are available in the data analysis toolkit. We don't have to rest all our expectations on Excel alone.

R

R is the next tool in our toolkit in this course. R is an integrated suite of software facilities for data manipulation, calculation, and graphical display.

What's great about R is it is Open Source Software, meaning it is freely available for anyone to use (but more about this later). R is one of my favourite tools to analyse data as it has some of the best data manipulation, data visualisation and result reporting capabilities. We will cover R in more detail in module 2 and 3, so stay tuned for the exciting journey that lies ahead!

Tableau

Tableau is the last tool we will use to our advantage in our toolkit. It is free for students and we will end our journey with this tool by exploring it in more detail in module 4.

Where does the data come from?

The next step in the data analyst journey would be to ask yourself Where does the data come from? Are we using all possible data sources available to draw insights from?

If you are combining several different data sources, it might be good to think about setting up ETL service (once again, more on this later).

Quality

When thinking about where the data is sourced from, we need to consider its quality. In the case of data from medical tests, we could investigate the percentage of false positive to false negative tests we receive, meaning we would look at how sensitive the tests are.

Relevance

This step can also look at when the data was recording, is your data relevant to today or is it outdated?

Basic data cleaning

We could consider some basic data cleaning in this step, by adding in more fields (e.g. unique variables) to create new cumulative variables to investigate. Checking the quality of your data is exceptionally important, because herein lies the foundation for analysis. If the data is incorrect, you will draw conclusions that are incorrect.



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Analysing the data

Now we get to analysing the data:

1. Decide on technique to apply
2. List limitations, uncertainties, unknowns and
3. Identify missing data

Reporting and presenting

- Consider carry over learnings (what do they already know)
- Who is your audience (i.e. what are their needs, technical skills and how much time do they have)?
- Format: Report or presentation
- Type of visualisation (more on this in lesson 2)

What is interesting about what we know about data analysis and how we utilise this skill today, is that before computers, the US census of 1880 took over 7 years to process the collected data. A machine capable of systematically processing data recorded on punch cards was luckily invented. This cut down the time needed to analyse this data so that in the 1890 census it only took 18 months to do the job.

Another turning point was reached when relational databases came into being in the 1980s which allowed us to analyse data on demand through Structured Query Language (SQL). This sped up the process of using data to draw insights for everyday use and gradually more computational techniques led us to what we know as data analysis today (i.e. being able to use live data to draw conclusions).

The concepts of data analysis will always have its roots in statistical analysis, but as computational techniques evolve, the two become more integrated. This requires a data analyst to understand the statistical techniques involved, but also to be able to utilise these computational techniques to extract insights from in hopefully less than 18 months! Changing technologies ensure that data analysis is ever changing and a lifelong journey of learning.

Is data analysis for me?

Data analysis is for everyone! It's a skill that you can use to better understand data in your everyday life but it's also a skill that can be used by a business to draw insights from the data they have available and use those insights to make better data driven decisions to aid their business decisions.

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Why should I learn this skill?

Data analysis has become synonymous with problem solving. It can impact the way a business serves its customers. Because of the growing skills gap, analytical skills such as data analysis, has become integral to not just technological companies, but diverse industries such as insurance, marketing, product management, customer experience and many more.

For businesses to stay competitive, it has become essential to analyse data and find meaningful insights to use for better decision making in the business world. Choosing to follow a data analysis path, places you at the forefront of the decision-making process in the company.

Pursuing a career in data analysis allows you to choose between a variety of industries and the high demand for the skill means that this is a valuable role. Analytics is everywhere which means that new opportunities in this sector are constantly cropping up. It's a hugely exciting time to be a part of this industry and start a career in analytics. There is no doubt that analytics will continue to be a huge part of enterprises in the years to come, so without delay, let's get you started on the road to analysing some data!

Introduction to data

What is data?

- Data are observations or measurements (unprocessed or processed) represented as text, numbers, or multimedia.
- e.g. The height of Mount Everest is 'data'. This piece of information informs the mountaineer how they can prepare to ascend the mountain.

What is dataset?

A dataset is a structured collection of data generally associated with a unique body of work. A database is an organised collection of data stored as multiple datasets, that are generally stored and accessed electronically from a computer system that allows the data to be easily accessed, manipulated, and updated. We manipulate and investigate data from the dataset that is stored in the database.

But where does all this data come from? Data is information that is collected through observation and can be **qualitative or quantitative**.

Qualitative data

- Non-numerical data
- Collected interviews, focus groups
- Open-ended questions are used
- Subjective (not objective)
- Helps us to generate a hypothesis (more on this later in this module)
- This data is used to answer the 'why'

Quantitative data

- Numerical (meaning it is based on numbers)
- The sample size is usually bigger
- Statistics are used to draw conclusions
- Objective information (no subjective)
- Questions are close ended
- Used to validate a hypothesis
- Upon further investigation, we will subject our quantitative data to being discrete or continuous (more on this in lesson 2)

Where is data used?

As mentioned previously, we use data in a vast array of industries to help make better decisions. Scientific research, businesses, finance, governance, non-profits use data, in fact any organisation you can think of uses data. As a data analyst, you will analyse and report findings back to the industries that have collected data.

Data storage

Data analysts will use databases to access the data. Remember that a database is a collection of information that is organised and easily accessible, managed and updated.

Data can be stored in some of the following files

- Excel or comma separated value files (.xlsx, .csv)
- Text files
- XML files
- JSON files
- Data can also be stored in databases

Types of databases

Relational database

A relational database is a collection of information that organises data points with defined relationships for easy access. In the relational database model, the data structures including data tables, indexes and views which remain separate from the physical storage, allowing administrators to edit the physical data storage without affecting the logical data structure.

Distributed database

A distributed database is a collection of multiple interconnected databases, which are spread physically across various locations that communicate via a computer network.

Object-oriented database

An object-oriented database is a database that subscribes to a model with information represented by objects.

Graph database

A graph database is a database designed to treat the relationships between data as equally important to the data itself. It is intended to hold data without constricting it to a pre-defined model.

Where can you get data from now?

In the 1980's Richard Stallman started what can be seen today as the Open Source movement.

Part of this movement includes:

- Open data in line with open movements like open source, open hardware to make these tools and data free and easy to access.
- This is important because data grows exponentially every day. The hypothesis is that if there are restrictions, businesses and governments will not be able to become more data driven in their approach.

Sources of Open Data

World bank data

As a repository of the world's most comprehensive data regarding what's happening in different countries across the world, World Bank Open Data is a vital source of Open Data. It also provides access to other datasets as well which are mentioned in the data catalogue.

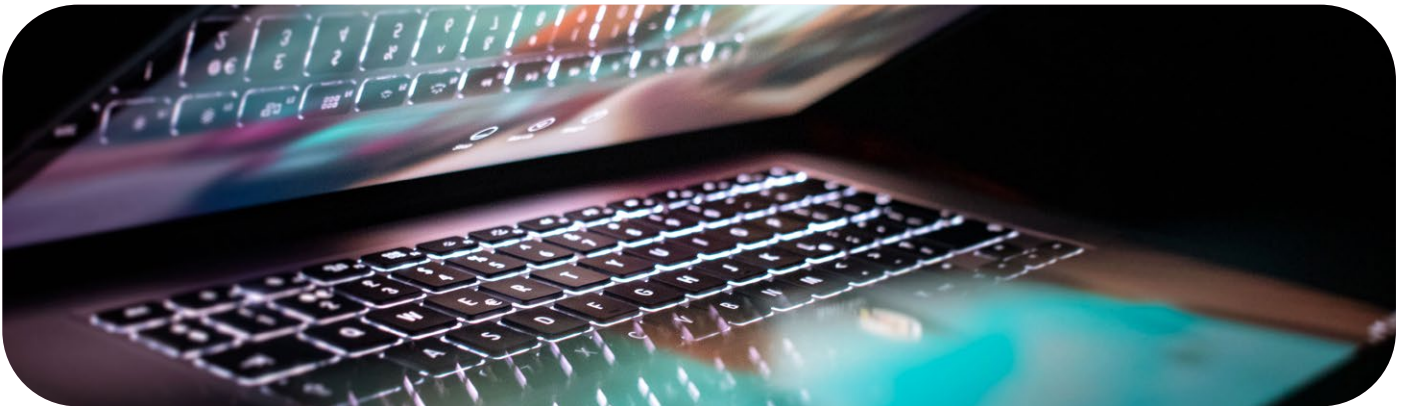
- 3000+ datasets
- Allows you to download in different formats

World Health Organization (WHO)

- Health specific data for 194 member states
- **Categories include:** Mortality or burden of disease, child nutrition, child health, HIV/Aids, many more
- Available to download in Excel format

Google Public Data Explorer

- Public interest data
- Play around with data by creating visualisations and share the link
- Makes data available from different sources. E.g. World Bank, US bureau of labor statistics, International Monetary Fund (IMF), etc.

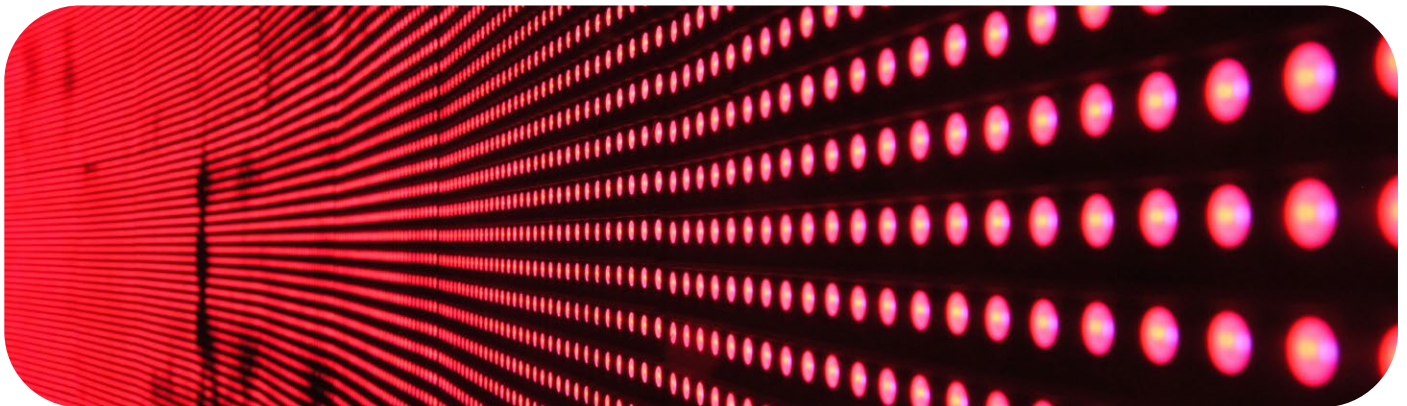


Kaggle

- Variety of datasets
- Encourages publishers to share data in an accessible format
- Encourages cross collaboration with other data analysts, scientists, and engineers
- Promotes competitions to solve challenges
- Users publish code snippets

Titanic dataset

On April 15, 1912, the largest passenger liner ever made collided with an iceberg during her maiden voyage. When the Titanic sank it killed 1502 out of 2224 passengers and crew. This sensational tragedy shocked the international community and led to better safety regulations for ships. One of the reasons that the shipwreck resulted in such loss of life was that there were not enough lifeboats for the passengers and crew. Although there was some element of luck involved in surviving the sinking, some groups of people were more likely to survive than others. Kaggle contains an open source dataset that holds information about the survival of Titanic passengers. Prior to your lesson you must head over to Kaggle and download the dataset to follow along with me for the final topic of importing and cleaning this dataset with the help of Power Query in Excel.



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Importing and cleaning data

We will utilise Excel as a tool for this step as well as the Titanic dataset. We will also do some initial 'logical checks' on the dataset to make sure that it is 'clean' before we start with lesson 2.

How do we get the data into Excel?

Manually

For this method you will manually copy-paste data into an Excel spreadsheet. The problem with this process is that it is slow, repetitive and error prone.

Macros and Visual Basics for Applications (VBAs)

Macros and VBAs help us to automate the importing process. Our Excel course offers more on this if you are interested. This method requires some programming knowledge and requires you to spend some time maintaining.

Power Query

We will use the Power Query Extract, Transform, Load (ETL) tool available in Excel to import our dataset.

Why are we using Power Query?

Power Query is a Business Intelligence tool available in Excel which helps us to manipulate data. It can connect to different data sources, combine, and transform them. With Power Query, you can reuse queries (i.e. set up a query once and refresh it when new data becomes available). No coding knowledge is required, but you can use M-code ('miscellaneous function') if you want to write your own.

What can Power Query do?

Think of power query as your extract, transform and load tool in Excel. It allows you to:

- **Extract:** Use Power Query to discover and connect to a variety of data sources.
- **Transform:** Transform the extracted data by, for example, combining or refining it.
- **Load:** Share the transformed data.

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Conclusion

Data truly is everywhere and understanding it will not only help you understand the world we live in but it will help you make sense of it too and open a variety of opportunities for you and your career. Remember that the best way to master data analysis, is to practise what we did during the lesson. You can download open source datasets and play around with them if you are interested in other sources of information, like finance or medicine.

In lesson 2 we will start exploring data in further detail. We will learn more about different data types, different ways data can be graphically represented and learn more about some basic descriptive statistics. Throughout lesson 2, we will practise what we learn on the Titanic dataset and dive deeper into the sea of data!

References

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